

Philosophy of Science
Prof. James W. McAllister

Lecture 3

Scientific Laws and Induction

Motivation of this lecture

- To make you more effective and reflective users of two conceptual tools
 - Scientific law, induction
 - Crucial in scientific research and applications of knowledge
- To give you deeper awareness of the strengths of the nomothetic approach
 - In social and human sciences too

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Overview of this lecture

- Nomothetic approach: a reminder
- Concept of scientific law
 - A glance at physics
 - Examples of laws in social and human sciences
- Induction: a technique for detecting regularities and formulating laws
 - David Hume's problem of induction
 - Pragmatic vindication of inductivism

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Reading for this lecture

- Yuri V. Balashov, "Laws of Nature"
 - In *Encyclopedia of Science and Religion*, edited by J. Wentzel Vrede van Huyssteen
 - Macmillan Reference, 2003; pp. 508–513
- Ahmed Jamal Anwer, "Some Approaches to the Problem of Induction"
 - *Indian Philosophical Quarterly* 22 (1995), pp. 247–258
- Access the readings via Brightspace
 - Click the "Available here" link

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Reminder: nomothetic approach

- What does the nomothetic approach consist in?
 - Identifying regularities in the world
 - Formulating laws and generalisations to describe these regularities
 - Deriving explanations of observed outcomes from those laws and generalisations
- Contrast with the idiographic approach

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Laws in physics

- Laws of nature...
 - are fundamental principles of physics
 - identify important regularities of the universe
- Characteristics of laws:
 - Mathematical equations
 - Concise and simple, often elegant
 - Universal in scope
- Two examples follow

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Laws in physics: example 1

- Inverse-square law of gravitation
 - Isaac Newton, 1642–1727

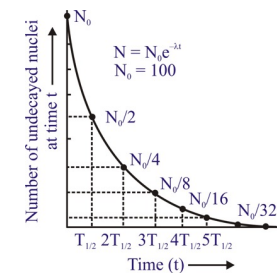
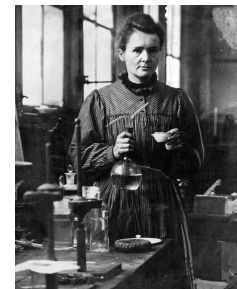
$$F = Gm_1m_2/r^2$$

- Expresses...
 - the force F due to gravity between two masses m_1 and m_2 separated by distance r
- Example: force between sun and earth

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Laws in physics: example 2

- Law of radioactive decay: $N(t) = N_0e^{-\lambda t}$
 - Marie Skłodowska Curie, 1867–1934



Decay curve for a radioactive element

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Extension to other sciences

- Nomothetic approach...
 - can be applied also in social and human sciences
- Presupposition: human affairs exhibit significant regularities
 - Describable by universal statements
 - Largely independent of context
 - Somewhat similar to regularities in physics
- Laws in social and human sciences

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Four examples

- Linguistics and cognitive science
 - Berlin and Kay's principle of acquisition of basic colour terms
- Political science
 - Duverger's law of electoral systems
- Economics
 - Tinbergen's gravity model of international trade
- International relations
 - Democratic peace theory

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Basic colour terms

- As languages evolve, they acquire basic colour terms in a universal sequence
 - Brent Berlin and Paul Kay, *Basic Color Terms: Their Universality and Evolution*. University of California Press, 1969
 - Data from twenty languages
- Universalist hypothesis
 - Colour cognition is an innate, physiological process
 - Independent of cultural factors

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Basic colour terms: details

- Languages acquire colour terms in the following order, say Berlin and Kay:
 - 1-2. Terms for dark-cool and light-warm (e.g. English "black", "white")—all languages have these
 3. Red
 4. Either green or yellow
 5. Both green and yellow
 6. Blue
 7. Brown
 8. Purple, pink, orange, or grey

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Basic colour terms: discussion

- If Berlin and Kay are right, then...
 - This aspect of human cognition is a physiological universal
 - Specifics of linguistic communities are less influential than we normally assume in human sciences
 - There is a biological framework underlying cultural manifestations
- Other scientists have questioned...
 - Berlin and Kay's interpretation of the data

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Duverger's law

- Plurality-rule electoral systems tend to yield two-party parliaments
 - Example: first past the post—UK, US
- Proportional representation tends to yield multipartyism
 - Netherlands, Belgium, Sweden
- Dynamic of voting systems explains degree of political fragmentation
 - Maurice Duverger, French sociologist and member of the European parliament, 1950s

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Gravity model of trade

- Volume of bilateral trade V between two countries is determined by:
 - Their economic size, m_1 and m_2
 - Geographical distance between them, d
- $$V = Gm_1m_2/d$$
- Jan Tinbergen, Dutch economist, 1962
 - Similarity with Newton's law of gravity
 - Suggestion: underlying physical principles determine regularities of human behaviour

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Democratic peace theory

- Democratic states do not wage war with one another
 - Closest thing to an empirical law in international relations
 - Inspired by Immanuel Kant's essay, *Zum ewigen Frieden* ("Perpetual Peace"), 1795
- Critical discussion
 - How to define "democracy", "war"
 - Historical exceptions and doubtful cases
 - How to explain the regularity

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Reflections on these examples

- These laws embody the view that...
 - We can say significant things about human affairs without regard for cultural setting
 - There is substantial universality or regularity underlying human affairs...
 - determined in some cases by biological, physiological or structural constraints
- Contrast with the idiographic approach in social and human sciences

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Laws: regularities—or more?

- Do laws specify just what *is* the case...
 - or what in some sense *must be* the case?
- Regularity view of laws
 - World: just events—no necessary relations
 - Generalisations describe empirical regularities; important ones are called “law”
- Necessity view of laws
 - World: necessary and accidental regularities
 - Laws describe the necessary regularities

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How do we discover laws?

- Top-down approach
 - Deduction from theoretical first principles
- Bottom-up approach
 - Generalisation from observed regularities
- Induction is the method for...
 - detecting regularities in empirical data
 - formulating generalisations and empirical laws
- To induction we now turn!

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What is induction?

- Inductivist strategy works as follows:
 - The observer notes any initial indication of a regularity in a few cases:
$$x_1 \text{ is an } A \text{ and a } B$$
$$x_2 \text{ is an } A \text{ and a } B$$
 - Then, the observer hypothesises that the regularity holds for all cases:
$$\text{All } A\text{s are } B\text{s}$$
- Result: inductive generalisation

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What is induction? Example

- The observer notes an initial indication of a regularity in swans:

x_1 is a swan and is white

x_2 is a swan and is white

- Then, the observer hypothesises that the regularity holds for all swans, including swans not yet observed:

All swans are white

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Induction vs. deduction

- Deduction goes from general to particular:

All As are Bs
 x_1 is an A

 x_1 is a B

- Induction goes from particular to general:

x_1 is an A and a B
 x_2 is an A and a B

...

All As are Bs

← Ranges over some,
but not all, As

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Logical features of induction

- Inductive inferences go beyond observations performed
 - Induction is ampliative, unlike deduction
- Inductive arguments are not logically compelling
 - An inductive generalisation may be disconfirmed by the next case: a black swan
- That's why induction is...
 - productive and useful in research...
 - but risky!

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David Hume, 1711–1776



- Historian and philosopher
- Worked in Edinburgh
- Protagonist of the Scottish Enlightenment
- Left: Allan Ramsay, *David Hume* (1766), Scottish National Portrait Gallery, Edinburgh

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Hume's problem of induction

- How can we justify the conclusion of an inductive inference?
- There are only two possible ways to justify induction, reasoned Hume:
 - Induction—but this leads to infinite regress
 - Deduction—but this is too narrow to justify ampliative conclusions
- So neither way of justifying induction is effective

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Hume's conclusion

- There is no logical justification for inductive generalisations
 - At most, induction must be seen as a psychological tendency of the human mind
- Philosophical responses to Hume
 - Some writers have reiterated Hume's view that induction cannot be justified. Example: Karl R. Popper—see lecture 5
 - Others have proposed vindications of induction. Example: pragmatic warrant

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Inductivism: pragmatic warrant

- Originators:
 - Hans Reichenbach, Richard B. Braithwaite
- We cannot be sure that the world contains any universal regularities
- If the world does contain regularities...
 - then induction is at least as quick as any other strategy to detect these regularities
 - Why? Because induction consists in noting any initial regularity, and hypothesising that it holds indefinitely

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Pragmatic warrant: discussion

- Offers a pragmatic justification...
 - for using induction in cognition and science
- Hume's original problem still stands
 - We still lack justification of individual inductive conclusions, e.g. "All swans are white"
- However, we have shifted the focus to...
 - justification of an inductivist cognitive strategy

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Recapitulation

- Nomothetic approach
 - Typical of the natural sciences, applicable also to human affairs
- Concept of scientific law
 - Examples in social and human sciences
- Induction as a technique for...
 - detecting regularities, formulating laws
- Hume's problem of induction
 - And a pragmatic justification of inductivism